

TEACHER MASTER KEY: ALL 3 HONORS CHALLENGES

Quick-Reference Answers, Coordinates & Debrief Solutions

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Challenge 1: Ship & Drone Kinematic Rendezvous

Dock Origin ($x = 0$)

Graphed Points: Ship: (0s, 300m), (10s, 360m), (20s, 420m), (30s, 480m), (40s, 540m), (50s, 600m), (55s, 630m). Drone: (0 to 20s at 0m), (30s, 180m), (40s, 360m), (50s, 540m), (55s, 630m).

- **Rendezvous Coordinates:** $t = 55.0$ seconds, $x = 630.0$ meters from dock.
- **Slopes:** Ship = $+6.0$ m/s • Drone (after $t=20s$) = $(540 - 0)/(50 - 20) = +18.0$ m/s.
- **Drone Flight Duration:** $55\text{ s} - 20\text{ s} = 35.0$ seconds in the air.
- **Battery Safety Audit:** Round-trip = $630\text{ m} + 630\text{ m} = 1,260\text{ m}$. Battery limit is $1,200\text{ m} \rightarrow$ **FAIL** (Will crash 60 m short of dock).

Challenge 2: Tactical Highway Surveillance Intercept

Piecewise Motion

Graphed Points: Target: (0s, 0m) $\rightarrow$ (20s, 300m) [Leg 1] $\rightarrow$ (50s, 300m) [Leg 2, flat line] $\rightarrow$ (90s, 1300m) [Leg 3].

- **Rest Stop Meaning:** Zero slope = zero velocity (car is stationary / not moving).
- **Patrol Line A (Catch at Rest Stop at $t=50s$):** Slope = $(300\text{ m} - 0\text{ m}) / (50\text{ s} - 30\text{ s}) = 300 / 20 = 15.0$ m/s.
- **Patrol Line B (Catch at Highway End at $t=90s$):** Slope = $(1,300\text{ m} - 0\text{ m}) / (90\text{ s} - 30\text{ s}) = 1,300 / 60 = 21.7$ m/s.
- **Speed Limit Constraint (Cap = 20.0 m/s):** Strategy A requires $15.0\text{ m/s} \leq 20.0\text{ m/s}$ (**VIABLE**). Strategy B requires $21.7\text{ m/s} > 20.0\text{ m/s}$ (**TOO FAST / IMPOSSIBLE**).

Challenge 3: Alpine Mountain Rescue Relay

Distance vs. Displacement

Graphed Points: Rover: (0s, 0m) $\rightarrow$ (120s, 1800m). Drone: (0s, 0m) $\rightarrow$ (25s, 500m) $\rightarrow$ (60s, 500m) [pause] $\rightarrow$ (85s, 1000m).

- **Arrival Times:** Drone reaches $1,000\text{ m}$ at $t = 85$ seconds • Rover reaches $1,800\text{ m}$ at $t = 120$ seconds.
- **Margin of Victory:** Drone Beta wins by 35.0 seconds ($120\text{ s} - 85\text{ s}$).
- **Path Distance vs. Displacement:** Rover travels winding switchback road ($1,800\text{ m}$); drone flies straight-line vector ($1,000\text{ m}$).
- **Maximum Allowable Cooldown:** Rover takes 120 s . Drone flies for 50 s total. $120\text{ s} - 50\text{ s} = 70.0$ seconds max pause.